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The role of neural activity in the migration and differentiation of enteric neuron precursors

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Abstract

Background *As they migrate through the developing gut, a sub-population of enteric neural crest-derived cells (ENCCs) begins to differentiate into neurons. The early appearance of neurons raises the possibility that electrical activity and neurotransmitter release could influence the migration or differentiation of ENNCs.* ***Methods*** *The appearance of neuronal sub-types in the gut of embryonic mice was examined using immunohistochemistry. The effects of blocking various forms of neural activity on ENCC migration and neuronal differentiation were examined using explants of cultured embryonic gut.* ***Key Results*** *Nerve fibers were present in close apposition to many ENCCs. Commencing at E11.5, neuronal nitric oxide synthase (nNOS), calbindin and IK_{Ca} channel immunoreactivities were shown by sub-populations of enteric neurons. In cultured explants of embryonic gut, tetrodotoxin (TTX, an inhibitor of action potential generation), nitro-L-arginine (NOLA, an inhibitor of nitric oxide synthesis) and clotrimazole (an IK_{Ca} channel blocker) did not affect the rate of ENCC migration, but tetanus toxin (an inhibitor of SNARE-mediated vesicle fusion) significantly impaired ENCC migration as previously reported. In explants of E11.5 and E12.5 hindgut grown in the presence of TTX or tetanus toxin there was a decrease in the number nNOS+ neurons close to the migratory wavefront, but no significant difference in the proportion of all ENCC that expressed the pan-neuronal marker, Hu.* ***Conclusions & Inferences*** *(i) Some enteric neuron sub-types are present very early during the development of the enteric nervous system. (ii) The rate of differentiation of some sub-types of enteric neurons appears to be influenced by TTX- and tetanus toxin-sensitive mechanisms.*

Keywords *differentiation, enteric nervous system, migration, neural activity, neural crest, neuronal nitric oxide synthase.*

In the developing central nervous system (CNS), electrical activity and spontaneous release of neurotransmitters influence neural proliferation, migration and neuronal differentiation, including neurotransmitter specification, neuronal morphology, and axon guidance.[1–6] For example, endogenous GABA and glutamate influence the speed of migration of neurons in a number of regions of the developing brain,[7–9] control neuroblast number in proliferative zones,[10] and regulate the neurotransmitter phenotype of neurons in the developing spinal cord.[11] It has been proposed that electrical activity acts in partnership with genetic programs during early neural development.[3]

The enteric nervous system (ENS) is a large network of neurons and glia within the wall of the gastrointestinal tract that controls many aspects of intestinal function.[12] The ENS arises predominantly from vagal neural crest cells that emigrate from the caudal hindbrain during embryonic development.[13–15] In mice, enteric neural crest-derived cells (ENCCs) enter the foregut around E9.5, and by E10, a sub-population (10–20%) starts to express pan-neuronal markers including Hu, neurofilament-M, peripherin, neuron-specific enolase, Tuj1 and PGP9.5.[16–21] Many of these immature enteric neurons have neurites[21,22] that project caudally in close association with the migrating ENCCs.[19] The early appearance of neurons, and their close relationship with migrating ENCCs, raises the possibility that electrical activity and neurotransmitter

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release from immature neurons could influence the migration, proliferation, differentiation or neurotransmitter specification of ENCCs.[23]

Botulinum neurotoxin A and tetanus neurotoxin (TeNT), which inhibit SNARE-mediated vesicle fusion, delay ENCC migration in explants of embryonic mouse gut.[24] Furthermore, a recent study has shown that pharmacological or electrical field stimulation-induced depolarization of enteric neurons derived by dissociation of E15 rat gut and culturing for 2 weeks increased the proportions of neurons expressing tyrosine hydroxylase (TH) or vasoactive intestine peptide (VIP).[25] Thus it appears that the migration of ENCCs and the phenotype of cultured developing enteric neurons can be regulated by neuronal activity. However, little else is known about the role of electrical activity and neurotransmitter release in the development of the ENS.

In the current study we show that sub-populations of neurons containing neuronal nitric oxide synthase (nNOS) and calbindin, and nerve fibers expressing the intermediate conductance, Ca^{2+}-dependent, potassium channel (IK_{Ca}), can be identified in the gut of E11.5 mice. The effects of blocking various forms of neural activity on ENCC migration and neuronal differentiation in explants of embryonic gut cultured for 48 h were examined. Enteric neural crest-derived cell migration was unaffected by pharmacological inhibition of sodium-dependent action potentials, NOS or the IK_{Ca} channel, but inhibition of SNARE-mediated vesicle fusion impaired ENCC migration. Pharmacological inhibition of sodium-dependent action potentials using tetrodotoxin (TTX) and inhibition of SNARE function using TeNT significantly reduced the number of nNOS neurons close to the migratory wavefront. Thus, TTX- and tetanus toxin-sensitive mechanisms appear to play a role in the differentiation of some sub-types of enteric neurons.

MATERIALS AND METHODS

Animals

Ret$^{TGM/+}$ mice were generated by inserting cDNA encoding tau-EGFP-myc (TGM) into the first exon of the *Ret* gene.[26] E11.5 and E12.5 *Ret*$^{TGM/+}$ heterozygote embryos were obtained by plug-mating male *Ret*$^{TGM/+}$ mice with female wild-type mice. Midday of the day at which a plug was found was designated as embryonic day 0.5 (E0.5). Pregnant females were killed by cervical dislocation and embryos decapitated prior to dissection. All mice were bred on a C57BL6J background. Wild-type C57BL6J mice were also used for some immunohistochemical studies. All procedures were approved by the University of Melbourne Anatomy & Cell Biology, Neuroscience, Pathology, Pharmacology, and Physiology Animal Ethics Committee (Ethics ID: 0809127).

Catenary (suspension) organ culture

Explants of embryonic gut were dissected and mounted across a 'V' cut in filter paper supports.[27] At E11.5, the mid plus hindgut was cultured as a single explant.[28] At E12.5, the middle region of the small intestine, and the entire colon, were grown separately on filter paper supports. The explants were grown for 48 h in Terasaki wells containing 20 μL of culture medium (DMEM/F12 with 20 mM L-glutamine, 6 mg mL^{-1} penicillin/streptomycin, and 10% fetal bovine serum).

Pharmacological inhibitors

Tetrodotoxin (1 μM; Alomone Labs, Jerusalem, Israel), nitro-L-arginine (NOLA, 100 μM; Sigma), TeNT (1 μg mL^{-1}; Sigma) or clotrimazole (10 μM; Sigma, St Louis, MO, USA) were added to the culture medium. Tetrodotoxin, NOLA and tetanus toxin were dissolved in distilled H_2O, and clotrimazole was dissolved in DMSO prior to addition to the culture medium. Control explants were grown in culture medium containing the same volume of the relevant vehicle as used with the pharmacological inhibitors. The effectiveness of the drugs was tested using physiological experiments: TTX (1 μM) abolished and NOLA (100 μM) significantly increased the frequency of colonic migrating motor complexes (CMMCs) in preparations of adult mouse colon as described previously.[29] Preliminary experiments with 50 ng mL^{-1} tetanus toxin[24] showed that this concentration of tetanus toxin did not significantly affect CMMCs in the mouse colon or contractions of adult mouse vas deferens induced by electrical field stimulation. However, 1 μg mL^{-1} tetanus toxin abolished CMMCs and contractions of adult mouse vas deferens induced by electrical stimulation and so 1 μg mL^{-1} tetanus toxin was used in all experiments in the current study. Intracellular recordings showed that clotrimazole (10 μM) abolished the after-hyperpolarization (AH) in AH neurons of adult guinea-pig, as reported previously for mouse enteric neurons.[30]

For both the migration and differentiation assays using E11.5 mice, the embryos of each litter were screened to identify those containing GFP+ cells in the stomach (*Ret*$^{TGM/+}$ embryos). One half of the *Ret*$^{TGM/+}$ embryos from each litter were used as controls and the other half were drug-treated. Thus, there were separate control experiments for each drug treatment. For the experiments to examine the effects of tetanus toxin and TTX on neuronal differentiation on small intestine and colon from E12.5 mice, one-third of the embryos were controls, one-third exposed to tetanus toxin and one-third to TTX.

Immunohistochemistry

Tissue to be processed for IK_{Ca} channel immunoreactivity was fixed in 2% formaldehyde plus 0.2% picric acid in 0.1M phosphate buffer and was processed for immunohistochemistry as described previously.[30] All other tissues were fixed in 4% formaldehyde in 0.1M phosphate buffer and processed for immunohistochemistry as described previously.[31] A list of primary antibodies used is shown in Table 1. Secondary antisera used were: donkey antihuman Texas Red (1 : 100; Jackson Immunobiologicals, West Grove, PA, USA), donkey antiguinea pig biotin followed by streptavidin Alexa 488 (1 : 100; Molecular Probes, Mulgrave, VIC, Australia), donkey antisheep FITC (1 : 100; Jackson), horse antimouse biotin (Vector, Burlingame, CA, USA) followed by streptavidin Alexa 594 (1 : 100; Molecular Probes) or donkey antirabbit Alexa 594 (1 : 100; Molecular Probes). Preparations were imaged using a Zeiss Pascal confocal laser scanning microscope.

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Table 1 Primary antisera used

	Host	Source or reference	Concentration
Calbindin	Rabbit	Swant	1 : 1600
Calretinin	Goat	Swant	1 : 100
CGRP	Goat	Biogenesis	1 : 1000
ChAT	Goat	Chemicon	1 : 100
GFP	Goat	Rockland	1 : 400
Hu	Human	Gift from Miles Epstein[37]	1 : 2000
IK_{Ca} channel	Rabbit	[30]	1 : 2000
nNOS	Sheep	Gift from Piers Emson[75,76]	1 : 2000
Substance P	Guinea-pig	Chemicon	1 : 1000
Tuj1	Mouse	Covance research products, USA	1 : 2000
VAChT	Goat	Chemicon	1 : 1000
Vasoactive intestinal peptide	Rabbit	Chemicon	1 : 1000

CGRP, calcitonin gene-related peptide; ChAT, choline acetyltransferase; nNOS, neuronal nitric oxide synthase; VAChT, vesicular acetylcholine transporter.

Migration analysis in cultured explants of gut

The distance migrated by GFP+ ENCCs during the 48 h culture period was determined by measuring the distance between the ileocecal junction and the most caudal ENCC in the postcecal hindgut as described previously.[28] Measurements were made using ImageJ software (NIH, Bethesda, MD, USA).

Neuron differentiation analysis in cultured explants of gut

Preparations were imaged using a ×63 objective lens. Explants of E11.5 gut were imaged in the (i) mid-midgut and (ii) postcecal hindgut. Explants of E12.5 gut were imaged in the (i) mid-small intestine, (ii) proximal colon, and (iii) distal colon. The number of GFP+ cells, Hu+ cells and nNOS+ cells were counted in single optical sections in each region, and the data compared using ANOVAs on the raw data. All results are presented as mean ± SEM.

RESULTS

Developmental appearance of pan-neuronal and neuronal sub-type specific markers

In E10.5 mice, ENCCs have colonized the stomach and rostral midgut, and by E11.5, the most caudal ENCCs are entering the postcecal hindgut.[32,33] All ENCCs within the developing gut of E10.5-E13.5 $Ret^{TGM/+}$ mice express EGFP.[34] As reported previously, in E10.5-E12.5 $Ret^{TGM/+}$ mice, a sub-population of Ret+ ENCCs expressed the pan-neuronal markers, Hu, Tuj1 (Fig. 1-A,B) and neurofilament-M, and Tuj1+ processes were observed to extend from many of the Tuj1+ cell bodies (Fig. 1B). Tuj1+ and neurofilament-M+ nerve fibers were present in close apposition to many of the GFP+ cells, including those at the migratory wavefront (Fig. 1B). In regions already colonized by ENCCs, many of the nerve fibers formed prominent bundles oriented in the long axis of the gut (Fig. 1A).

We examined whether immunoreactivities to the enteric neuron subtype-specific markers, nNOS, vasoactive intestinal peptide (VIP), substance P, neuropeptide Y (NPY), calcitonin gene-related peptide (CGRP), choline acetyltransferase (ChAT), the vesicular acetylcholine transporter (VAChT), calbindin, calretinin and the IK_{Ca} channel were present in the gut of E10.5, E11.5 and E12.5 mice. None of the neuron subtype markers could be detected in the gut of E10.5 mice. In the gut of E11.5 and E12.5 mice, nNOS, calbindin and IK_{Ca} channel immunoreactivities were observed, but no specific immunostaining was observed with the other antibodies tested. A sub-population of the GFP+/Hu+ cell bodies expressed nNOS and/or calbindin (Fig. 2A,B). In E11.5 mice, when the wavefront of GFP+ cells is entering the postcecal hindgut, the most caudal nNOS+ neurons were present in the cecum (Fig. 2A'). The most caudal calbindin+ neurons were usually slightly rostral to the most caudal nNOS+ neuron, and were present in the caudal midgut (Fig. 2A). Most nNOS+ neurons did not show any calbindin immunostaining and *vice versa* (Fig. 2B), but occasional nNOS+/calbindin+ neurons were also observed. The nNOS+ fibers formed a network in close association with many of the GFP+ cells (Fig. 1C), and some nNOS+ fibers extended caudally beyond the most caudal nNOS+ cell body (Fig. 1D). However, nNOS+ fibers were not observed associated with GFP+ cells at the extreme wavefront, in the postcecal hindgut. Although many of the calbindin+ neurons possessed short processes (Fig. 2B), calbindin+ fibers were rarely observed. The IK_{Ca} channel immunoreactivity was shown by a sub-population of Tuj1+ nerve fibers in the rostral midgut (Fig. 1E), but not in the caudal midgut, cecum or postcecal hindgut of E11.5 mice. The IK_{Ca} channel immunostaining appeared as puncta along a sub-population of the Tuj1+ nerve fibers (Fig. 1E). The IK_{Ca} channel+ fibers were closely associated with GFP+ ENCCs. Rare IK_{Ca} channel+ cell bodies were also observed in the rostral midgut of some preparations.

Effect of blockers of sodium dependent action potentials, neurotransmitter release, neuronal nitric oxide synthase and the IK_{Ca} channel on migration of neural crest-derived cells

As a network of nerve fibers is in close apposition with migrating ENCCs, we examined whether ENCC migration was affected by pharmacological inhibitors

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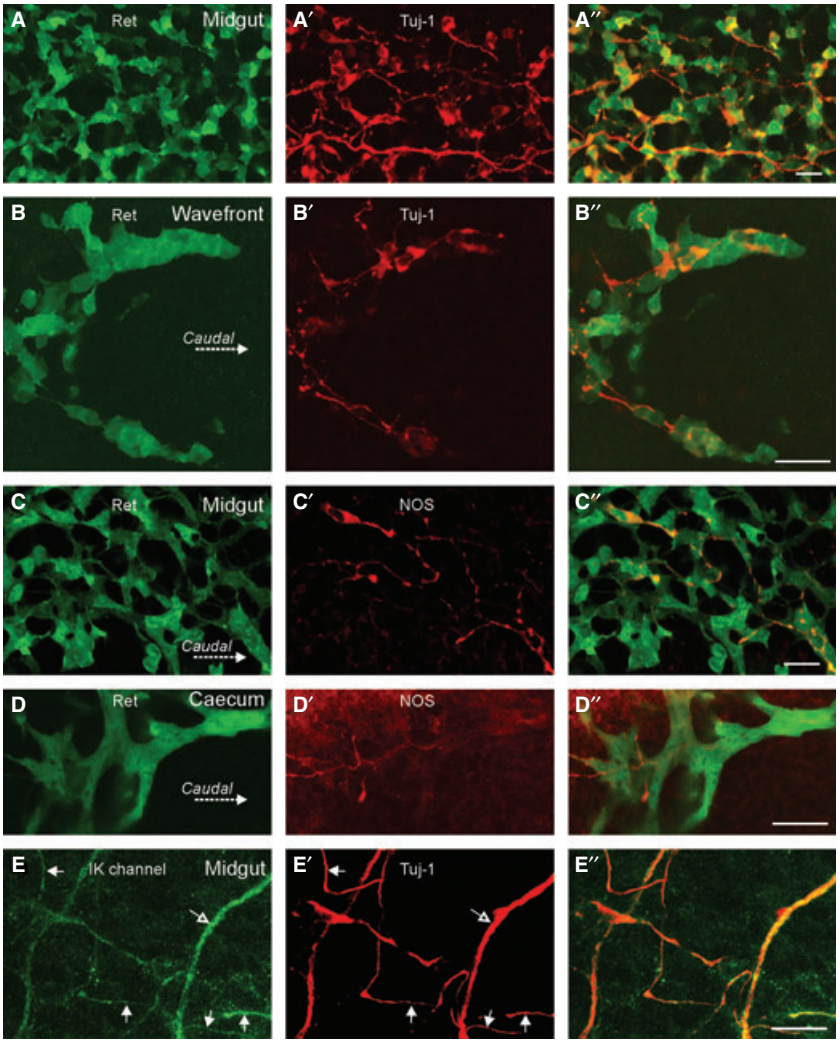

Figure 1 Wholemount preparations of E11.5 gut. (A, B) Tuj1 immunoreactive fibers (red) form close appositions with Ret+ (GFP+) ENCC from *Ret*^{TGM} mice in both the midgut (A–A″) and at the migratory wavefront in the cecum (B–B″). In the midgut (A′), nerve fibers formed prominent bundles oriented in the long axis of the gut. (C, D) Varicose nNOS+ fibers are present in close association with Ret+ ENCC in both the midgut (C–C″) and close to the migratory wavefront at the entrance to the cecum (D–D″). nNOS+ fibers could not, however, be detected in association with the most caudal Ret+ cells in the postcecal colon. (E) Individual IK_{Ca} channel immunoreactive fibers (*filled arrows*) as well as fiber bundles (*open arrow*), in the rostral midgut of wild-type mice. The IK_{Ca} channel immunoreactive fibers are a sub-population of the Tuj1+ fibers (E′). IK_{Ca} channel immunoreactivity could not be detected in the distal midgut, cecum or postcecal hindgut. Scale bar: 50 μm (applies to all).

of a variety of types of neural activity. Tetrodotoxin was used to block action potentials dependent on inward Na^+ currents, tetanus toxin to block SNARE-mediated vesicle fusion and thus synaptic vesicle exocytosis, NOLA to block nitric oxide synthase (NOS) activity and clotrimazole to block IK_{Ca} channels. Control physiological experiments using adult gut showed that all drugs were active (see Materials and Methods).

Explants of midgut plus hindgut from E11.5 *Ret*$^{TGM/+}$ embryos were set up as catenary (suspension) organ cultures[28] and grown with and without the pharmacological inhibitors for 48 h. At the beginning of the culture period, GFP+ ENCCs were just entering the postcecal hindgut. After 48 h in control culture conditions, GFP+ ENCCs had colonized most of the postcecal hindgut (Fig. 3A). The distance between the ileocecal junction and the most caudal GFP+ ENCC was measured in control medium (containing vehicle only) or in medium containing the various pharmacological inhibitors. The distance migrated by ENCC was not significantly affected by TTX (1312 ± 79 μm *vs* 1372 ± 62 μm in controls; mean ± SEM, $P = 0.55$, one-way ANOVA), NOLA (1555 ± 164 μm *vs* 1415 ± 203 μm in controls, $P = 0.61$) or clotrimazole (1544 ± 135 *vs* 1481 ± 105 μm in controls; $P = 0.71$) after 48 h in culture (Fig. 3). In contrast, tetanus toxin (1 μg mL^{-1}) significantly reduced the distance that ENCCs migrated (1163 ± 66 μm *vs* 1504 ± 76 μm in controls; $P = 0.006$, one-way ANOVA; Fig. 3). Proliferation is essential for the migration of ENCCs.[35] To determine if tetanus toxin retards ENCC migration by reducing the number of ENCCs, the density of GFP+ ENCCs was determined in the postcecal hindgut of control and tetanus toxin-treated explants. There was no significant difference in the density of GFP+ cells in the

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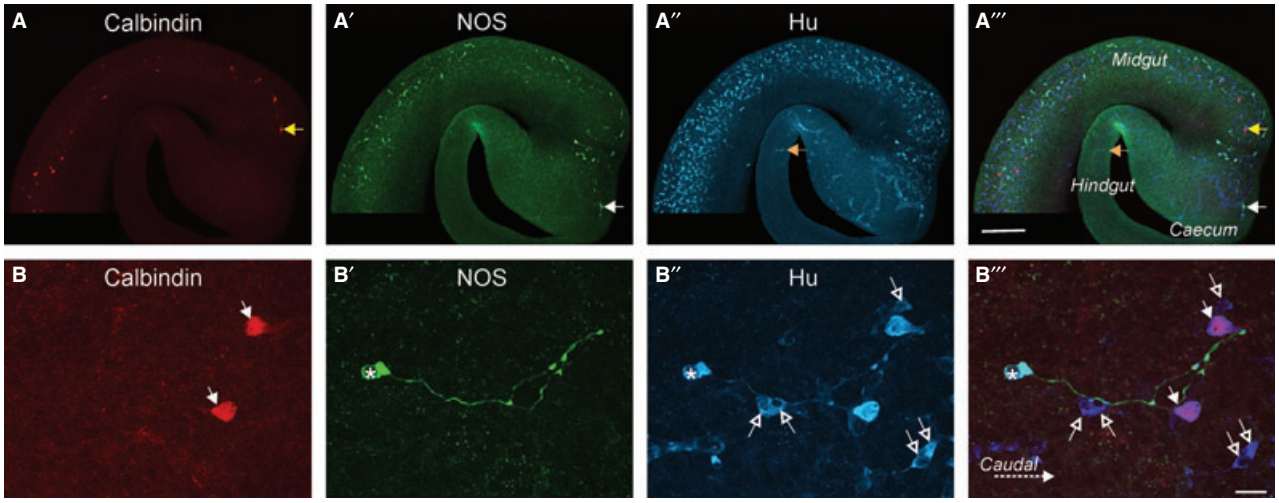

Figure 2 Wholemount preparations of mid- plus hindgut from an E11.5 mouse immunostained for calbindin (red), nNOS (green) and Hu (red). (A) Low magnification image. The most caudal calbindin+ neuron is in the caudal midgut (yellow arrow), the most caudal nNOS+ cell is in the cecum (*white arrow*) and the most caudal Hu+ cell is in the postcecal hindgut (*orange arrow*). Scale bar: 200 μm. (B) Higher magnification image of calbindin (*arrows*), nNOS (*asterisk*) and Hu neurons in the midgut. Calbindin and nNOS were present in largely non-overlapping populations of neurons. The nNOS neuron has a caudally projecting process (B'), but processes could not be detected arising from calbindin neurons (B). Many Hu+ neurons (*open arrows*) do not contain calbindin or nNOS. Scale bar: 20 μm.

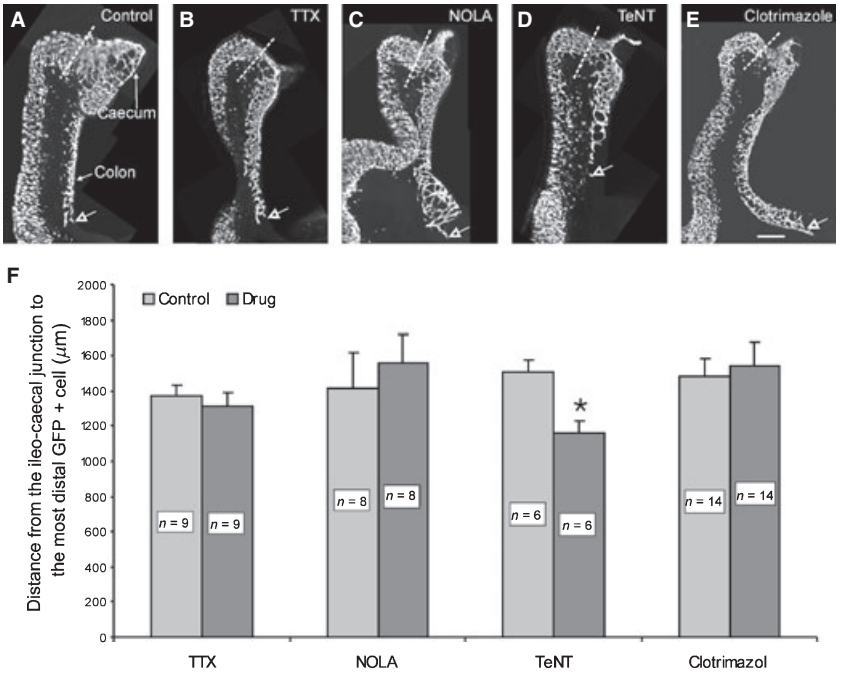

Figure 3 Effects of blocking various forms of neural activity on ENCC migration. (A–E) Explants of mid- plus hindgut from E11.5 Ret^{TGM} mice were grown in culture for 48 h under control conditions or in the presence of tetrodotoxin (TTX), nitro-L-arginine (NOLA), tetanus toxin (TeNT) or clotimazole, and then fixed and processed for GFP immunostaining. The distance from the ileo-cecal junction (*dotted line*) to the most caudal cell (*open arrows*) was measured. The location of the most caudal cell (*open arrows*) was in the distal colon in all explants except those grown in the presence of tetanus toxin (D). Scale bar: 200 μm. (F) Quantification of the distance from the ileo-cecal junction to the most caudal GFP+ ENCC. Only explants grown in the presence of tetanus toxin showed a significant difference from controls (*t*-test, $P < 0.05$).

postcecal hindgut (control: 54 ± 8 GFP+ cells per 10 000 μm^2, $n = 4$. Tetanus toxin-treated: 66 ± 5 GFP+ cells per 10 000 μm^2, $n = 3$; $P = 0.30$, Student's *t*-test), suggesting that the retardation in migration induced by tetanus toxin is not due to an indirect effect on the numbers of ENCCs.

There was also no detectable effect of any of the drugs on the overall appearance of the network of ENCCs.

Effects of blockers of sodium-dependent action potentials, neurotransmitter release, nitric oxide synthase and the IK_{Ca} channel on differentiation of Hu+ and nNOS+ neurons

To examine whether differentiation of ENCCs into Hu+ and nNOS+ neurons was affected by blocking a variety of types of neural activity, explants of E11.5 and E12.5 $Ret^{TGM/+}$ gut were cultured in the presence of

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TTX, NOLA, tetanus toxin or clotimazole for 48 h and then processed for immunohistochemistry using antisera against GFP, Hu and nNOS. Two regions were examined for E11.5 preparations: the mid-midgut and the rostral hindgut, caudal to the cecum. Three regions were examined for E12.5 preparations: the mid small intestine, proximal colon, and distal colon. For each area, the percentages of (i) GFP+ cells that were Hu+, and (ii) Hu+ cells that were nNOS+, were determined.

For E11.5 cultures, the percentage of GFP+ cells expressing Hu was similar in both regions, and there was no significant difference in the GFP+ : Hu+ ratio in the presence of any of the drugs (Fig. 4A,C; percentage of GFP cells that were Hu+: (i) Midgut: control = 43 ± 3%, clotrimazole = 46 ± 5%, $P = 0.60$, one-way ANOVA; control = 26 ± 3%, NOLA = 29 ± 2%, $P = 0.33$; control = 24 ± 3%, tetanus toxin = 23 ± 4%, $P = 0.78$; control = 30 ± 4%, TTX = 29 ± 5%; $P = 0.93$. (ii) Postcecal hindgut: control = 42 ± 5%, clotrimazole = 41 ± 5%, $P = 0.92$; control = 26 ± 3%, NOLA = 22 ± 3%, $P = 0.36$; control = 38 ± 4%, tetanus toxin = 45 ± 5%, $P = 0.26$; control = 26 ± 5%, TTX = 27 ± 5%, $P = 0.91$). There was a higher proportion of Hu+ cells that were nNOS+ in the midgut than the rostral hindgut (58 ± 3% *vs* 30 ± 3% respectively). Tetanus toxin and TTX both significantly reduced the percentage of Hu+ neurons that showed nNOS immunoreactivity in the rostral hindgut (9 ± 4% of Hu+ cells were nNOS+ in the presence of tetanus toxin *vs* 31 ± 8% in controls, $P = 0.036$; 11 ± 5% of Hu+ cells were nNOS+ in the presence of TTX *vs* 33 ± 7% in controls, $P = 0.026$, one-way ANOVA), but not the midgut (Fig. 4B,C; Midgut: control = 48 ± 8%, tetanus toxin = 35 ± 9%, $P = 0.29$; control = 60 ± 8%, TTX = 49 ± 6%, $P = 0.35$). Inhibition of IK_{Ca} channel activity using clotrimazole, and nitric oxide production using NOLA had no effect on the percentage of Hu+ neurons that were nNOS+ in the midgut or hindgut (Fig. 4B; Midgut: control = 59 ± 5%, clotrimazole = 48 ± 5%, $P = 0.12$; control = 63 ± 5%, NOLA = 57 ± 8%, $P = 0.52$. Rostral hindgut: control = 23 ± 6%, clotrimazole = 30 ± 11%, $P = 0.61$; control = 34 ± 5%, NOLA = 43 ± 5%, $P = 0.21$). The differences in the control values between different groups of experiments could reflect slight age differences between experiments, differences between batches of tissue culture medium components and/or differences in immunostaining between experiments.

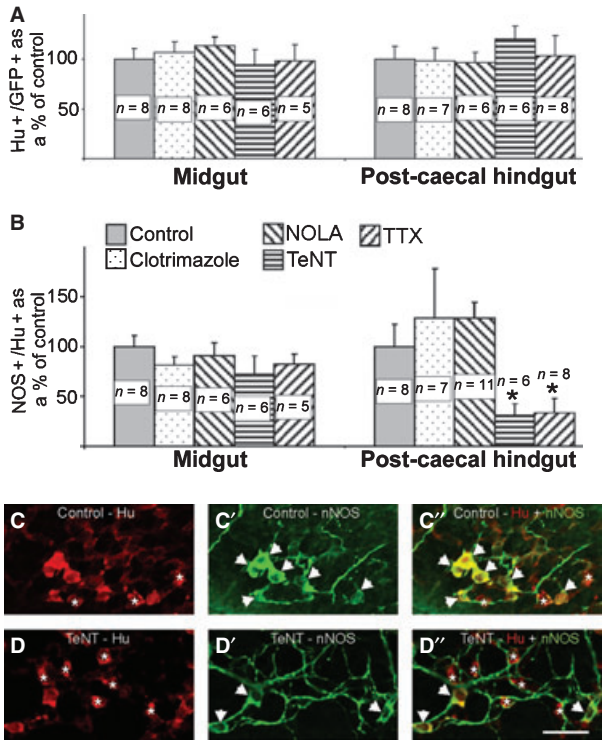

Figure 4 Effects of blocking various forms of neural activity on differentiation of Hu+ and nNOS+ neurons. Explants of mid- plus hindgut from E11.5 Ret^{TGM} mice were grown in culture for 48 h under control conditions or in the presence of tetrodotoxin (TTX), nitro-L-arginine (NOLA), tetanus toxin (TeNT) or clotimazole, and then fixed and processed for immunohistochemistry using antisera to GFP, nNOS and Hu. (A) The proportion of GFP+ ENCC that was Hu+ was unaffected by any of the drugs in both the midgut and the rostral hindgut. (B) The proportion of Hu+ neurons that was nNOS+ was unaffected by any of the drugs in the midgut but in the rostral hindgut, the percentage of Hu+ neurons that was nNOS+ was lower following exposure to tetanus toxin (31 ± 11% of the percentage of Hu+ neurons that are nNOS+ of control cultures) or TTX (33 ± 15% of control cultures; one-way ANOVA, $P < 0.05$). (C, D) Micrographs of Hu and nNOS staining in the postcecal hindgut of E11.5 preparations cultured under control conditions for 48 h (C–C″) and preparations cultured in the presence of tetanus toxin (D–D″). In preparations grown in the presence of tetanus toxin (D″), there were fewer nNOS+ cells (*arrows*) and more Hu+/nNOS-negative cells (*asterisks*) than in control preparations (C″). Scale bar: 50 μm (applies to all).

A reduction in the number of nNOS+ neurons in the hindgut, but not midgut, of E11.5 mice following exposure to TTX or tetanus toxin suggests that the timing of exposure to the drugs is important. We therefore cultured explants of small intestine and colon from E12.5 mice in the presence of tetanus toxin or TTX. Both tetanus toxin and TTX produced a significant reduction in the proportion of Hu+ neurons that also expressed nNOS in the E12.5 distal colon when cultured for 48 h (16 ± 3% of Hu+ neurons were nNOS+ in the presence of tetanus toxin, and 13 ± 3% in the presence of TTX, compared with 27 ± 3% in control cultures; $P = 0.024$ and 0.004, one-way ANOVA followed by Holm-Sidak *post hoc* analysis; Fig. 5). There was no obvious effect of TTX or tetanus toxin on the number of nNOS+ fibers in the hindgut (Fig. 4C',D'). Moreover, neither tetanus toxin nor TTX produced a significant change in the proportion of

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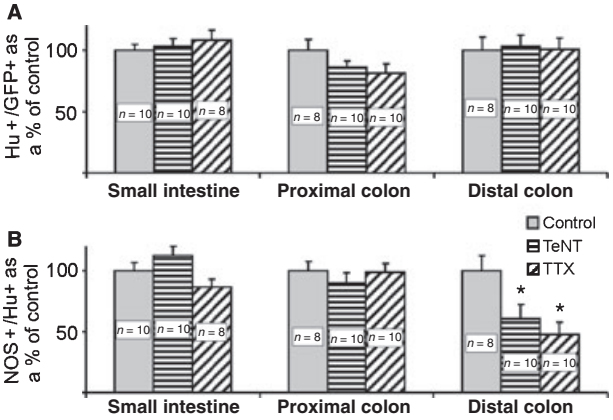

Figure 5 Effects of blocking voltage-gated sodium channels using tetrodotoxin (TTX) and SNARE-dependent vesicle exocytosis using tetanus toxin on the differentiation of Hu+ and nNOS+ neurons in gut explants from E12.5 Ret^{TGM} mice. Preparations of E12.5 mid-small intestine and the entire colon were cultured for 48 h, fixed and processed for immunohistochemistry using antisera against GFP, Hu and nNOS. (A) The proportion of GFP+ ENCCs that differentiated into Hu+ neurons was not affected in any of the three regions examined: small intestine, proximal colon, and distal colon. (B) The proportion of Hu+ neurons that were also nNOS+ was decreased in the distal colon by both drugs (tetanus toxin: 61 ± 10% of control cultures; TTX: 48 ± 12% of control cultures; one-way ANOVA, $P < 0.05$, one-way ANOVA, $P < 0.05$), but was unaffected in the small intestine and proximal colon.

nNOS+ cell bodies in the proximal colon or mid-small intestine (proximal colon: control = 53 ± 4% of Hu+ cells were also nNOS+, tetanus toxin = 48 ± 4%, TTX = 52 ± 4%, $P = 0.60$; mid-small intestine: control = 38 ± 3%, tetanus toxin = 42 ± 2%, $P = 0.20$, TTX = 33 ± 3%, $P = 0.19$). The proportion of GFP+ cells expressing Hu was also not affected by tetanus toxin or TTX in any of the three regions of E12.5 gut cultured for 48 h (distal colon: control = 33 ± 4% of GFP+ cells were also Hu+, tetanus toxin = 34 ± 3%, TTX = 33 ± 3%, $P = 0.98$; proximal colon: control = 53 ± 5%, tetanus toxin = 46 ± 4%, TTX = 43 ± 3%; $P = 0.20$; mid-small intestine: control = 58 ± 3%, tetanus toxin = 60 ± 5%, TTX = 63 ± 4%; $P = 0.71$).

Immunostaining of cultured explants with the calbindin antibody resulted in high background staining and it was not possible to distinguish unequivocally specific calbindin immunostained cells. Thus, the effects of neural activity on the differentiation of calbindin neurons could not be examined.

DISCUSSION

Previous studies have shown that neuronal differentiation commences shortly after ENCC first enter the gut.[16–18,36–39] The main findings of this study are (i) processes of immature enteric neurons are in close apposition to many ENCC, including those at the migratory wavefront, (ii) commencing at E11.5, sub-populations of immature neurons express the sub-type specific markers, nNOS, calbindin or the IK_{Ca} channel, (iii) TTX (blocker of voltage-dependent Na^+ currents of action potentials), NOLA (blocker of nitric oxide synthesis) and clotrimazole (blocker of IK_{Ca} channels) had no effect on ENCC migration in explants of gut from E11.5 mice, but tetanus toxin (blocker of SNARE-mediated vesicle fusion) retarded ENCC migration as reported previously,[24] and (iv) exposure to TTX or tetanus toxin caused a delay in the differentiation of nNOS+ neurons close to the migratory wavefront.

Differentiation of enteric neuronal subtypes

The mature ENS is made up of a wide variety of different neuronal types that vary in neurochemistry, projections, and electrophysiology.[40–44] Although immature neurons expressing pan-neuronal markers are present in the mouse gut from E10.5,[16,18,45] neuron sub-type specific markers first appear later.[23,46–48] Enteric neural crest-derived cells expressing *Cart* (cocaine- and amphetamine-regulated transcript) can be detected in the gut of E11.5 mice,[49] and previous studies reported the presence of nNOS neurons in the gut of E12.5 mice.[19,47,50] In the current study, we showed nNOS neurons can be detected in the gut of E11.5 mice, as well as neurons expressing calbindin and nerve fibers expressing the IK_{Ca} channel. The nNOS neurons also appear early during the development of the zebrafish ENS.[51,52] In the ENS of mature mice, most nNOS neurons are inhibitory neurons to the circular muscle but they also include some interneurons.[40,43] The IK_{Ca} channel and calbindin are present in intrinsic sensory neurons plus some other neuronal sub-populations.[30,43] None of the other markers of enteric neurons investigated (ChAT, VAChT, calretinin, VIP, substance P and NPY) were detectable at E11.5. Vasoactive intestine peptide is present in nNOS neurons in the small and large intestines of adult mice.[40,43] Although the current study showed that nNOS+ neurons were present at E11.5, VIP is not observed until E14.[47,53] Likewise, calbindin was observed at E11.5, but markers that are co-localized with calbindin in the adult, such as ChAT, calretinin and CGRP,[40,43] are not detectable until later in embryonic development.[47,54] Thus there is a staggered appearance of markers of the mature phenotype in individual neuronal subtypes. Previous studies have shown that the earliest expression of NPY is at E13-13.5, VIP and substance P at E14-14.5, and CGRP at E17-17.5.[47,53]

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In embryonic rats and mice, the sub-population of ENCCs that first expresses pan-neuronal markers also transiently expresses the catecholamine synthesizing enzyme, TH and contain catecholamines,[18,55–57] raising the possibility that catecholamines might influence ENCC migration and differentiaton. There are at least 10 known catecholamine receptors, including dopaminergic and α and β adrenergic receptors. Future studies are required to determine which catecholamine receptors are expressed in the developing gut and at which times they are first expressed, so that pharmacological antagonists can be used to determine whether catecholamines play a role in the development of enteric neurons.

Neural activity and ENCC migration

In some regions of the developing mammalian CNS and the insect nervous system, spontaneous neural activity, especially the release of neurotransmitters, plays an important role in the control of neuronal migration.[8,9,58–61] As there is a network of nerve fibers in close apposition with migrating ENCCs, we examined whether ENCC migration was affected by blocking a variety of types of neural activity. We found that inhibition of voltage-dependent Na^+ channels, NOS or the IK_{Ca} channel had no detectable effect on the rate of migration along explants of gut from E11.5 mice grown in culture. Mice that are haploinsufficient for *Ret* are more sensitive to the development of aganglionosis than wild-type mice following genetic[62] or environmental[22] manipulations. In the current study, $Ret^{TGM/+}$ mice were used and the migration of ENCCs was not significantly affected by TTX, NOLA or clotrimazole, even in this sensitized background.

In agreement with a previous study,[24] tetanus toxin did impair ENCC migration. Tetanus toxin is a blocker of SNARE-mediated vesicle fusion, including synaptic vesicle release. In addition to synaptobrevin-2, a key SNARE protein at neuronal synapses, tetanus toxin also cleaves cellubrevin, a ubiquitously expressed SNARE that is involved in general vesicle trafficking within cells.[63,64] Tetanus toxin impairs migration of some non-neuronal cells.[65,66] The tetanus toxin-induced retardation of ENCC migration could therefore be due to an effect on cell membrane trafficking and/or an effect on neurotransmitter release. Future studies are required to reveal whether tetanus toxin affects the migratory behavior of individual ENCCs, their interactions with each other, with the extracellular matrix and/or mesenchymal cells in intact segments of gut; for example, whether tetanus toxin affects the formation of lamellipodia, filopodia or leading processes. Finally, it should be noted that as TTX does not block all forms of neural activity, we still cannot rule out the possibility that ENCC migration might be influenced by some forms of neural activity.

Previous studies have shown the importance of nitric oxide signaling in neural migration in a number of invertebrates.[61] Furthermore, in the grasshopper embryo, nitric oxide has been found to promote neuronal migration in the developing ENS.[67,68] However, although nerve fibers containing nNOS are in close association with ENCC in the gut of E11.5 mice, blocking nitric oxide synthesis had no detectable effects on the rate of migration of ENCC in gut explants.

Neural activity and ENCC differentiation

A role for electrical activity in ENS differentiation has been identified in a recent study, in which the effects of neural activity on the differentiation of enteric neuronal subtypes in cultured neurons from the gut of E15 rats were examined.[25] Depolarization was found to increase the proportion of VIP neurons.[25]

In the current study we showed that the differentiation of Hu+ neurons in explants of E11.5 gut was unaffected following inhibition of voltage-dependent Na^+ channels, NOS, SNARE-mediated vesicle fusion or IK_{Ca} channel activity. In contrast, there was a delay in the appearance of nNOS+ neurons close to the migratory wavefront (in the E11.5 rostral hindgut and in the E12.5 distal colon) following inhibition of voltage-dependent Na^+ channel activity with TTX, or SNARE-dependent vesicle exocytosis with tetanus toxin for 48 h. The results of our study suggest (i) TTX- and tetanus toxin-sensitive signalling is present early during ENS development and influences the differentiation of the nNOS+ subtype of enteric neurons, and (ii) the timing of these signals is important. The data suggest that at E11.5, nNOS+ neuron specification has already taken place in the midgut, while nNOS precursors in the postcecal hindgut, which are close to the migratory wavefront, are responsive to environmental cues. Consistent with this, by E12.5, nNOS+ neuron specification has progressed to the proximal colon and nNOS neuron development is no longer affected by the application of pharmacological inhibitors of neural activity, but nNOS precursors in the distal colon are influenced by TTX and tetanus toxin. The importance of timing on neuron subtype differentiation has been demonstrated in the developing *Xenopus* nervous system, where neurotransmitters were found to influence the differentiation of separate subsets of neurons, but only

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during a few key days of development.[11] Although nNOS+ neurons are a sub-population of Hu+ neurons, there were no significant effects of TTX or tetanus toxin on the proportion of ENCC that expressed Hu. This might reflect the fact that nNOS+ neurons comprise only a small sub-population of Hu+ neurons in the hindgut at the developmental stages examined, and/or that TTX and tetanus toxin specifically perturb the timing of expression of nNOS, not neuronal differentiation in general.

The expression of neurotransmitter-related molecules including synthetic enzymes and transporters is controlled by transcription factors.[69,70] However, previous *in vitro* and *in vivo* studies have shown that the differentiation of sub-types of enteric neurons can also be influenced by GDNF family members,[71] neurotrophins,[72] the cell adhesion molecule L1,[73] bone morphogenetic proteins[74] and electrical activity.[25]

CONCLUSIONS

Pan-neuronal markers are expressed by a sub-population of ENCC as the gut is being colonized. This study has shown that TTX and tetanus toxin influence the rate of differentiation of a sub-population of enteric neurons in gut explants. Thus, at least some of the ENCCs expressing pan-neuronal markers are likely to signal via TTX and tetanus toxin-sensitive mechanisms. Electrophysiological studies are required to identify definitively when neural activity begins during the development of enteric neurons, and the nature of the electrical activity.

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REFERENCES

1 Borodinsky LN, Root CM, Cronin JA, Sann SB, Gu X, Spitzer NC. Activity-dependent homeostatic specification of transmitter expression in embryonic neurons. *Nature* 2004; **429**: 523–30.
2 Manent JB, Demarque M, Jorquera I *et al.* A noncanonical release of GABA and glutamate modulates neuronal migration. *J Neurosci* 2005; **25**: 4755–65.
3 Spitzer NC. Electrical activity in early neuronal development. *Nature* 2006; **444**: 707–12.
4 Weissman TA, Riquelme PA, Ivic L, Flint AC, Kriegstein AR. Calcium waves propagate through radial glial cells and modulate proliferation in the developing neocortex. *Neuron* 2004; **43**: 647–61.
5 Raven MA, Orton NC, Nassar H *et al.* Early afferent signaling in the outer plexiform layer regulates development of horizontal cell morphology. *J Comp Neurol* 2008; **506**: 745–58.
6 Zou D-J, Chesler A, Firestein S. How the olfactory bulb got its glomeruli: a just so story? *Nat Rev Neurosci* 2009; **10**: 611–8.
7 Bolteus AJ, Bordey A. GABA release and uptake regulate neuronal precursor migration in the postnatal subventricular zone. *J Neurosci* 2004; **24**: 7623–31.
8 Bless EP, Walker HJ, Yu KW *et al.* Live view of gonadotropin-releasing hormone containing neuron migration. *Endocrinology* 2005; **146**: 463–8.
9 Heng JI, Moonen G, Nguyen L. Neurotransmitters regulate cell migration in the telencephalon. *Eur J Neurosci* 2007; **26**: 537–46.
10 Platel J-C, Dave KA, Bordey Al. Control of neuroblast production and migration by converging GABA and glutamate signals in the postnatal forebrain. *J Physiol* 2008; **586**: 3739–43.
11 Root CM, Velazquez-Ulloa NA, Monsalve GC, Minakova E, Spitzer NC. Embryonically expressed GABA and glutamate drive electrical activity regulating neurotransmitter specification. *J Neurosci* 2008; **28**: 4777–84.
12 Furness JB. *The Enteric Nervous System*, Massachusetts, USA: Blackwell Publishing, 2006.
13 Le Douarin NM, Teillet MA. The migration of neural crest cells to the wall of the digestive tract in avian embryo. *J Embryol Exp Morphol* 1973; **30**: 31–48.
14 Yntema CL, Hammond WS. The origin of intrinsic ganglia of trunk viscera from vagal neural crest in the chick embryo. *J Comp Neurol* 1954; **101**: 515–41.
15 Burns AJ, Le Douarin NM. The sacral neural crest contributes neurons and glia to the post- umbilical gut: spatiotemporal analysis of the development of the enteric nervous system. *Development* 1998; **125**: 4335–47.
16 Baetge G, Gershon MD. Transient catecholaminergic (TC) cells in the vagus nerves and bowel of fetal mice: relationship to the development of enteric neurons. *Dev Biol* 1989; **132**: 189–211.
17 Baetge G, Schneider KA, Gershon MD. Development and persistence of catecholaminergic neurons in cultured explants of fetal murine vagus nerves and bowel. *Development* 1990; **110**: 689–701.
18 Young HM, Ciampoli D, Hsuan J, Canty AJ. Expression of ret-, p75(NTR)-, Phox2a-, Phox2b-, and tyrosine hydroxylase-immunoreactivity by undifferentiated neural crest-derived cells and different classes of enteric neurons in the embryonic mouse gut. *Dev Dyn* 1999; **216**: 137–52.
19 Young HM, Jones BR, McKeown SJ. The projections of early enteric neurons are influenced by the direction of neural crest cell migration. *J Neurosci* 2002; **22**: 6005–18.
20 Bondurand N, Natarajan D, Barlow A, Thapar N, Pachnis V. Maintenance of mammalian enteric nervous system progenitors by SOX10 and endothelin 3 signalling. *Development* 2006; **133**: 2075–86.
21 Barlow AJ, Wallace AS, Thapar N, Burns AJ. Critical numbers of neural crest cells are required in the path-

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ways from the neural tube to the foregut to ensure complete enteric nervous system formation. *Development* 2008; **135**: 1681–91.
22 Stewart AL, Young HM, Popoff M, Anderson RB. Effects of pharmacological inhibition of small GTPases on axon extension and migration of enteric neural crest-derived cells. *Dev Biol* 2007; **307**: 92–104.
23 Young HM. Functional development of the enteric nervous system – from migration to motility. *Neurogastroenterol Motil* 2008; **20**(Suppl. 1): 20–31.
24 Vohra BP, Tsuji K, Nagashimada M *et al.* Differential gene expression and functional analysis implicate novel mechanisms in enteric nervous system precursor migration and neuritogenesis. *Dev Biol* 2006; **298**: 259–71.
25 Chevalier J, Derkinderen P, Gomes P *et al.* Activity-dependent regulation of tyrosine hydroxylase expression in the enteric nervous system. *J Physiol* 2008; **586**: 1963–75.
26 Enomoto H, Crawford PA, Gorodinsky A, Heuckeroth RO, Johnson EM Jr, Milbrandt J. RET signaling is essential for migration, axonal growth and axon guidance of developing sympathetic neurons. *Development* 2001; **128**: 3963–74.
27 Hearn CJ, Young HM, Ciampoli D, Lomax AE, Newgreen D. Catenary cultures of embryonic gastrointestinal tract support organ morphogenesis, motility, neural crest cell migration, and cell differentiation. *Dev Dyn* 1999; **214**: 239–47.
28 Anderson RB, Turner KN, Nikonenko AG, Hemperly J, Schachner M, Young HM. The cell adhesion molecule L1 is required for chain migration of neural crest cells in the developing mouse gut. *Gastroenterology* 2006; **130**: 1221–32.
29 Roberts RR, Murphy JF, Young HM, Bornstein JC. Development of colonic motility in the neonatal mouse-studies using spatiotemporal maps. *Am J Physiol Gastrointest Liver Physiol* 2007; **292**: G930–8.
30 Neylon CB, Nurgali K, Hunne B *et al.* Intermediate-conductance calcium-activated potassium channels in enteric neurones of the mouse: pharmacological, molecular and immunochemical evidence for their role in mediating the slow afterhyperpolarization. *J Neurochem* 2004; **90**: 1414–22.
31 Anderson RB, Bergner AJ, Taniguchi M *et al.* Effects of different regions of the developing gut on the migration of enteric neural crest-derived cells: a role for Sema3A, but not Sema3F. *Dev Biol* 2007; **305**: 287–99.
32 Kapur RP, Yost C, Palmiter RD. A transgenic model for studying development of the enteric nervous system in normal and aganglionic mice. *Development* 1992; **116**: 167–75.
33 Young HM, Newgreen D. Enteric neural crest-derived cells: origin, identification, migration, and differentiation. *Anat Rec* 2001; **262**: 1–15.
34 Young HM, Bergner AJ, Anderson RB *et al.* Dynamics of neural crest-derived cell migration in the embryonic mouse gut. *Dev Biol* 2004; **270**: 455–73.
35 Simpson MJ, Landman KA, Hughes BD, Newgreen DF. Cell proliferation drives neural crest cell invasion of the intestine. *Dev Biol* 2007; **302**: 553–68.
36 Baetge G, Pintar JE, Gershon MD. Transiently catecholaminergic (TC) cells in the bowel of the fetal rat: precursors of noncatecholaminergic enteric neurons. *Dev Biol* 1990; **141**: 353–80.
37 Fairman CL, Clagett-Dame M, Lennon VA, Epstein ML. Appearance of neurons in the developing chick gut. *Dev Dyn* 1995; **204**: 192–201.
38 Newgreen DF, Hartley L. Extracellular matrix and adhesive molecules in the early development of the gut and its innervation in normal and spotting lethal rat embryos. *Acta Anat (Basel)* 1995; **154**: 243–60.
39 Conner PJ, Focke PJ, Noden DM, Epstein ML. Appearance of neurons and glia with respect to the wavefront during colonization of the avian gut by neural crest cells. *Dev Dyn* 2003; **226**: 91–8.
40 Sang Q, Young HM. Chemical coding of neurons in the myenteric plexus and external muscle of the small and large intestine of the mouse. *Cell Tissue Res* 1996; **284**: 39–53.
41 Furness JB. Types of neurons in the enteric nervous system. *J Auton Nerv Syst* 2000; **81**: 87–96.
42 Brookes SJ. Classes of enteric nerve cells in the guinea-pig small intestine. *Anat Rec* 2001; **262**: 58–70.
43 Qu ZD, Thacker M, Castelucci P, Bagyanszki M, Epstein ML, Furness JB. Immunohistochemical analysis of neuron types in the mouse small intestine. *Cell Tissue Res* 2008; **334**: 147–61.
44 Hao MM, Young HM. Development of enteric neuron diversity. *J Cell Mol Med* 2009; **13**: 1193–210.
45 Hao MM, Anderson RB, Kobayashi K, Whitington PM, Young HM. The migratory behavior of immature enteric neurons. *Dev Neurobiol* 2009; **69**: 22–35.
46 Rothman TP, Gershon MD. Phenotypic expression in the developing murine enteric nervous system. *J Neurosci* 1982; **2**: 381–93.
47 Branchek TA, Gershon MD. Time course of expression of neuropeptide Y, calcitonin gene-related peptide, and NADPH diaphorase activity in neurons of the developing murine bowel and the appearance of 5-hydroxytryptamine in mucosal enterochromaffin cells. *J Comp Neurol* 1989; **285**: 262–73.
48 Burns AJ, Pasricha PJ, Young HM. Enteric neural crest-derived cells and neural stem cells: biology and therapeutic potential. *Neurogastroenterol Motil* 2004; **16** (Suppl. 1): 3–7.
49 Heanue TA, Pachnis V. Expression profiling the developing mammalian enteric nervous system identifies marker and candidate Hirschsprung disease genes. *Proc Natl Acad Sci USA* 2006; **103**: 6919–24.
50 Woodward MN, Sidebotham EL, Connell MG *et al.* Analysis of the effects of endothelin-3 on the development of neural crest cells in the embryonic mouse gut. *J Pediatr Surg* 2003; **38**: 1322–8.
51 Holmberg A, Olsson C, Holmgren S. The effects of endogenous and exogenous nitric oxide on gut motility in zebrafish *Danio rerio* embryos and larvae. *J Exp Biol* 2006; **209**: 2472–9.
52 Olden T, Akhtar T, Beckman SA, Wallace KN. Differentiation of the zebrafish enteric nervous system and intestinal smooth muscle. *Genesis* 2008; **46**: 484–98.
53 Rothman TP, Nilaver G, Gershon MD. Colonization of the developing murine enteric nervous system and subsequent phenotypic expression by the precursors of peptidergic neruons. *J Comp Neurol* 1984; **225**: 13–23.
54 Young HM, Torihashi S, Ciampoli D, Sanders KM. Identification of neurons that express stem cell factor in the mouse small intestine. *Gastroenterology* 1998; **115**: 898–908.
55 Cochard P, Goldstein M, Black IB. Ontogenetic appearance and disappearance of tyrosine hydroxylase and catecholamines in the rat embryo.

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Proc Natl Acad Sci USA 1978; **75**: 2986–90.
56 Teitelman G, Baker H, Joh TH, Reis DJ. Appearance of catecholamine-synthesising enzymes during development of rat sympathetic nervous system: possible role of tissue environment. *Proc Natl Acad Sci USA* 1979; **76**: 509–13.
57 Gershon MD, Rothman TP, Joh TH, Teitelman GN. Transient and differential expression of aspects of the catecholaminergic phenotype during development of the fetal bowel of rats and mice. *J Neurosci* 1984; **4**: 2269–80.
58 Behar TN, Schaffner AE, Scott CA, Greene CL, Barker JL. GABA receptor antagonists modulate postmitotic cell migration in slice cultures of embryonic rat cortex. *Cereb Cortex* 2000; **10**: 899–909.
59 Behar TN, Smith SV, Kennedy RT, McKenzie JM, Maric I, Barker JL. GABA(B) receptors mediate motility signals for migrating embryonic cortical cells. *Cereb Cortex* 2001; **11**: 744–53.
60 Maric D, Liu QY, Maric I *et al.* GABA expression dominates neuronal lineage progression in the embryonic rat neocortex and facilitates neurite outgrowth via GABA(A) autoreceptor/Cl-channels. *J Neurosci* 2001; **21**: 2343–60.
61 Bicker G. STOP and GO with NO: nitric oxide as a regulator of cell motility in simple brains. *Bioessays* 2005; **27**: 495–505.
62 McCallion AS, Stames E, Conlon RA, Chakravarti A. Phenotype variation in two-locus mouse models of Hirschsprung disease: tissue-specific interaction between Ret and Ednrb. *Proc Natl Acad Sci USA* 2003; **100**: 1826–31.
63 Schiavo G, Benfenati F, Poulain B *et al.* Tetanus and botulinum-B neurotoxins block neurotransmitter release by proteolytic cleavage of synaptobrevin. *Nature* 1992; **359**: 832–5.
64 McMahon HT, Ushkaryov YA, Edelmann L *et al.* Cellubrevin is a ubiquitous tetanus-toxin substrate homologous to a putative synaptic vesicle fusion protein. *Nature* 1993; **364**: 346–9.
65 Tayeb MA, Skalski M, Cha MC, Kean MJ, Scaife M, Coppolino MG. Inhibition of SNARE-mediated membrane traffic impairs cell migration. *Exp Cell Res* 2005; **305**: 63–73.
66 Proux-Gillardeaux V, Gavard J, Irinopoulou T, Mege RM, Galli T. Tetanus neurotoxin-mediated cleavage of cellubrevin impairs epithelial cell migration and integrin-dependent cell adhesion. *Proc Natl Acad Sci USA* 2005; **102**: 6362–7.
67 Haase A, Bicker G. Nitric oxide and cyclic nucleotides are regulators of neuronal migration in an insect embryo. *Development* 2003; **130**: 3977–87.
68 Knipp S, Bicker G. Regulation of enteric neuron migration by the gaseous messenger molecules CO and NO. *Development* 2009; **136**: 85–93.
69 Goridis C, Rohrer H. Specification of catecholaminergic and serotonergic neurons. *Nat Rev Neurosci* 2002; **3**: 531–41.
70 Flames N, Hobert O. Gene regulatory logic of dopamine neuron differentiation. *Nature* 2009; **458**: 885–9.
71 Rossi J, Herzig KH, Voikar V, Hiltunen PH, Segerstrale M, Airaksinen MS. Alimentary tract innervation deficits and dysfunction in mice lacking GDNF family receptor alpha2. *J Clin Invest* 2003; **112**: 707–16.
72 Chalazonitis A, Pham TD, Rothman TP *et al.* Neurotrophin-3 is required for the survival-differentiation of subsets of developing enteric neurons. *J Neurosci* 2001; **21**: 5620–36.
73 Turner KN, Schachner M, Anderson RB. Cell adhesion molecule L1 affects the rate of differentiation of enteric neurons in the developing gut. *Dev Dyn* 2009; **238**: 708–15.
74 Chalazonitis A, Pham TD, Li Z *et al.* Bone morphogenetic protein regulation of enteric neuronal phenotypic diversity: relationship to timing of cell cycle exit. *J Comp Neurol* 2008; **509**: 474–92.
75 Norris PJ, Charles IG, Scorer CA, Emson PC. Studies on the localization and expression of nitric oxide synthase using histochemical techniques. *Histochem J* 1995; **27**: 745–56.
76 Young HM, Ciampoli D. Transient expression of neuronal nitric oxide synthase by neurons of the submucous plexus of the mouse small intestine. *Cell Tissue Res* 1998; **291**: 395–401.

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